

HTML5 – Day – 2 Hands on Activities – Vasundhara

Problem 1

Assessment Goal: Validate understanding of links and form elements.

Hands-on Tasks:

1. Create two HTML pages and link them using navigation links
2. Add a navigation bar with Home and Contact links
3. Create a registration form with:
 - o Name
 - o Email
 - o Password
 - o Gender (radio buttons)
 - o Course selection (dropdown)
4. Use required and placeholder attributes

Expected Outcome:

A multi-page website with working navigation and a basic form.

Source Code:

```
1  <!DOCTYPE html>
2  <html>
3  <head>
4  |   <title>Contact</title>
5  </head>
6  <body>
7
8  <nav>
9  |   <a href="index.html">Home</a> |
10 |   <a href="contact.html">Contact</a>
11 </nav>
12
13 <h2>Registration Form</h2>
14
15 <form>
16 |   <label>Name</label><br>
17 |   <input type="text" placeholder="Enter name" required><br><br>
18
19 |   <label>Email</label><br>
20 |   <input type="email" placeholder="Enter email" required><br><br>
21
22 |   <label>Password</label><br>
23 |   <input type="password" placeholder="Enter password" required><br><br>
24
25 |   <label>Gender</label><br>
26 |   <input type="radio" name="gender" value="Male" required> Male
27 |   <input type="radio" name="gender" value="Female" > Female
28 |   <br><br>
```

```

2 </html>
3
4 <body>
5 <form>
6
7   <label>Password</label><br>
8   <input type="password" placeholder="Enter password" required><br><br>
9
10  <label>Gender</label><br>
11  <input type="radio" name="gender" value="Male" required> Male
12  <input type="radio" name="gender" value="Female" > Female
13  <br><br>
14
15  <label>Course</label><br>
16  <input list="courses" placeholder="Select or type course" required>
17
18  <datalist id="courses">
19    <option value="Java">
20    <option value="Python">
21    <option value="Web Development">
22    <option value="Data Science">
23  </datalist>
24  <br><br>
25
26  <input type="submit" value="Register">
27
28 x</form>
29
30 </body>
31 </html>
32
33

```

Output:

[Home](#) | [Contact](#)

Registration Form

Name

Email

Password

Gender

☐ Male ☐ Female

Course

x

Explanation:

This task demonstrates the use of HTML `<nav>` and `<a>` elements to create navigation between multiple pages. It also uses basic form elements with `required` and `placeholder` attributes for client-side validation.

Problem 2: Simple Website Navigation System (Level-1)

Scenario

A small coaching institute wants a basic static website with multiple pages so students can easily navigate between sections like Home, Courses, and Contact.

Requirements

- Create 3 HTML pages:
 - index.html (Home)
 - courses.html
 - contact.html
- Use:
 - `<nav>` element to create a navigation menu
 - `<a>` anchor tags for navigation
- Navigation menu should appear on all pages
- Use relative paths to link pages
- Include at least:
 - One internal anchor link (`#section`)
 - One external link (e.g., institute's social media)

Technical Constraints

- Only HTML5
- No CSS or JavaScript
- Navigation must be semantic (`<nav>`)

Learning Outcome

You should be able to:

- Use `<nav>` and `<a>` elements
- Understand relative paths
- Build a simple multi-page navigation structure
- Users should move seamlessly between pages
- Code should be readable and well-indented
- Navigation links should work correctly

Source Code:

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>Home</title>
5 </head>
6 <body>
7
8 <nav>
9   <a href="index.html">Home</a> |
10  <a href="courses.html">Courses</a> |
11  <a href="contact.html">Contact</a>
12 </nav>
13
14 <h1>Welcome to Coaching Institute</h1>
15 <a href="#about">Go to About</a>
16
17 <h2 id="about">About Us</h2>
18 <p>This is a coaching institute.</p>
19
20 <a href="https://www.facebook.com">Visit Facebook</a>
21
22 </body>
23 </html>
24
25
26 <!DOCTYPE html>
27 <html>
28 <head>
29   <title>Courses</title>
30 </head>
31 <body>
32
33 <nav>
34   <a href="index.html">Home</a> |
35   <a href="courses.html">Courses</a>
36   <a href="contact.html">Contact</a>
37 </nav>
38
39 <h2>Our Courses</h2>
40 <ul>
41   <li>HTML</li>
42   <li>CSS</li>
43   <li>Java</li>
44 </ul>
45
46 </body>
47 </html>
48
49
50 <html>
51 <head>
52   <title>Contact</title>
53 </head>
54 <body>
55
56 <nav>
57   <a href="index.html">Home</a> |
58   <a href="courses.html">Courses</a>
59   <a href="contact.html">Contact</a>
60 </nav>
61
62 <h2>Contact</h2>
63 <p>Phone: 9999999999</p>
64
65 </body>
66 </html>
```

Output:

[Home](#) | [Courses](#) | [Contact](#)

Welcome to Coaching Institute

[Go to About](#)

About Us

This is a coaching institute.

[Visit Facebook](#)

Our Courses

- HTML
- CSS
- Java

Contact

Phone: 9999999999

Explanation:

This problem focuses on creating a semantic navigation system using <nav> and relative links between pages.

It ensures smooth user navigation across multiple HTML pages without using CSS or JavaScript.

Problem 3: Student Registration Form with Input Types (Level-1)

Scenario

An institute wants a basic student registration form with built-in validation to reduce incorrect submissions.

Requirements

Create an HTML webpage that includes:

Create a form with following input types:

- text (Name)
- email
- tel
- number (Age)
- date

Use **HTML5 built-in validation**:

- required
- min, max
- pattern (for phone number)

Add submit button

Technical Constraints

- HTML5 only
- No JavaScript validation
- Use proper <label> tags

Learning Outcome

You will be able to:

- Use advanced input types
- Apply built-in HTML5 validation
- Improve form usability without JavaScript
- Browser should show validation errors automatically
- Inputs should restrict invalid values
- Form should be user-friendly

Source Code:

```

1  <!DOCTYPE html>
2  <html>
3  <head>
4  |   <title>Student Registration</title>
5  </head>
6  <body>
7
8  <h2>Student Registration Form</h2>
9
10 <form>
11   <label>Name:</label><br>
12   <input type="text" required><br><br>
13
14   <label>Email:</label><br>
15   <input type="email" required><br><br>
16
17   <label>Phone:</label><br>
18   <input type="tel" pattern="[0-9]{10}" required><br><br>
19
20   <label>Age:</label><br>
21   <input type="number" min="18" max="60" required><br><br>
22
23   <label>Date of Birth:</label><br>
24   <input type="date" required><br><br>
25
26   <input type="submit" value="Submit">
27 </form>
28
29 </body>

```

Output:

Student Registration

Name:

Email:

Phone:

Age:

Date of Birth:

Submit

Explanation:

This form uses advanced HTML5 input types like email, tel, number, and date with built-in validation. Browser-level validation is achieved using attributes such as required, min, max, and pattern.

Problem 4: Multi-Page Website with Breadcrumb Navigation (Level-2)

Scenario

A training portal website has multiple levels of pages and users often get confused about where they are. The company wants breadcrumbs for better navigation.

Requirements

Pages:

- Home
- Courses
- Web Development
- HTML Course

Implement:

- `<nav>` for main menu
- Breadcrumb navigation using anchor tags

Example:

Home > Courses > Web Development > HTML

Use:

- **Hierarchical linking**
- **Relative paths**

Breadcrumb should update based on page hierarchy

Technical Constraints

- HTML5 only
- No CSS frameworks
- Semantic elements preferred

Learning Outcome

Learners should be able to:

- Breadcrumbs must reflect correct page structure
- Navigation must work without broken links
- Clear hierarchy in folder structure
- Design hierarchical navigation
- Implement breadcrumbs
- Understand real-world website structure

Source Code:

```
1  <!DOCTYPE html>
2  <html>
3  <head>
4  |   <title>HTML Course</title>
5  </head>
6  <body>
7
8  <nav>
9  |   <a href="index.html">Home</a> |
10 |   <a href="courses.html">Courses</a>
11 </nav>
12
13 <p>
14 |   <a href="index.html">Home</a> >
15 |   <a href="courses.html">Courses</a> >
16 |   <a href="webdev.html">Web Development</a> >
17 |   HTML
18 </p>
19
20 <h2>HTML Course</h2>
21 <p>Learn HTML from basics.</p>
22
23 </body>
24 </html>
25
```

Output:

[Home](#) | [Courses](#)

[Home](#) > [Courses](#) > [Web Development](#) > HTML

HTML Course

Learn HTML from basics.

Explanation:

This task implements breadcrumb navigation using anchor tags to represent page hierarchy.

It improves user orientation by showing the current page path using relative HTML links.

Problem 5: Course Enrollment Form with Advanced HTML Elements Scenario

An online learning platform wants to enhance user experience by showing course progress, skill levels, and suggested inputs while filling the form.

Requirements

Create an enrollment form using:

- `<datalist>` for course suggestions
- `<progress>` to show enrollment completion
- `<meter>` to indicate skill level (Beginner–Advanced)

Include modern input types:

- email
- range
- number

Use built-in validation

Technical Constraints

- HTML5 only
- No JavaScript or CSS
- Must use semantic elements

Learning Outcome

You will be able to:

- Use advanced HTML5 elements
- Enhance form UX without JavaScript
- Understand real-time data representation in HTML

Browser should display:

- Dropdown suggestions via `<datalist>`
- Progress bar updates (static value)
- Meter shows skill level visually

Clean and structured form layout

Source Code:

```
6    <body>
10   <form>
21       </datalist><br><br>
22
23       <label>Skill Level:</label><br>
24       <meter min="0" max="10" value="5"></meter><br><br>
25
26       <label>Progress:</label><br>
27       <progress value="60" max="100"></progress><br><br>
28
29       <label>Experience (years):</label><br>
30       <input type="range" min="0" max="10"><br><br>
31
32       <label>Age:</label><br>
33       <input type="number" min="18" required><br><br>
34
35       <input type="submit" value="Enroll">
36
37   </form>
38
39 </body>
40 </html>
41
```

```
1  <!DOCTYPE html>
2  <html>
3  <head>
4      <title>Course Enrollment</title>
5  </head>
6  <body>
7
8      <h2>Course Enrollment Form</h2>
9
10 <form>
11
12     <label>Email:</label><br>
13     <input type="email" required><br><br>
14
15     <label>Course:</label><br>
16     <input list="courses" required>
17 <datalist id="courses">
18     <option value="HTML">
19     <option value="CSS">
20     <option value="JavaScript">
21 </datalist><br><br>
22
23     <label>Skill Level:</label><br>
24     <meter min="0" max="10" value="5"></meter>
25
26     <label>Progress:</label><br>
27     <progress value="60" max="100"></progress>
28
29     <label>Experience (years):</label><br>
```

Output:

Course Enrollment Form

Email:

Course:

Skill Level:

Progress:

Experience (years):

Age:

Enroll

Explanation:

his problem uses advanced HTML5 elements like `<data list>`, `<progress>`, and `<meter>` for better user experience.

It demonstrates real-time visual feedback and input validation without JavaScript or CSS.